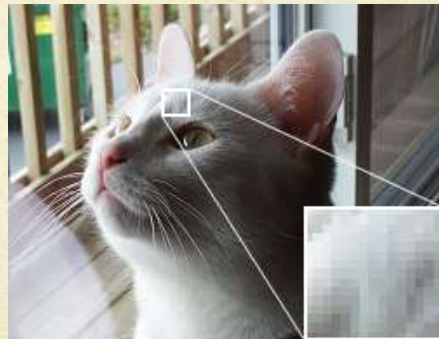
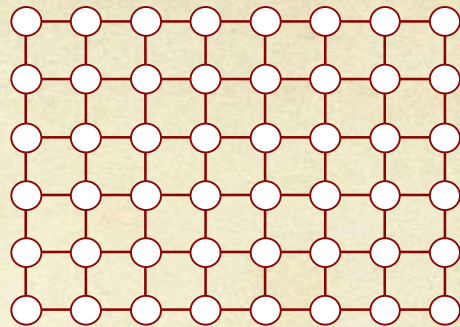


What does “image” mean for us?

- A 2D domain with samples at regular points (almost always a rectilinear grid)
 - Can have multiple values sampled per point
 - Meaning of samples depend on the application (red, green, blue, opacity, depth, etc.)
- Units also depend on the application
 - e.g., a computed int or float to be mapped to voltage needed for display of a pixel on a screen
 - e.g., as a physical measurement of incoming light (e.g., a camera pixel sensor)
- Introduction to sampling [demo](#)



What is a channel?

- A channel is all the samples of a particular type
- The red channel of an RGB image is an image containing just the red samples
- TV channels are tuned to different frequencies of electromagnetic waves
 - In a similar sense, RGB channels are “tuned” to different frequencies in the visible spectrum

Original RGB image
3 samples per pixel



Red channel
1 sample per pixel



Green channel
1 sample per pixel

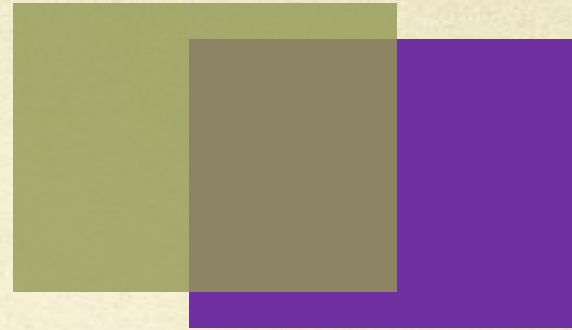


Blue channel
1 sample per pixel

2

The alpha channel

- In addition to the R, G, and B channels of an image, add a fourth channel called α (transparency/opacity/translucency)
- Alpha varies between 0 and 1
 - Value of 1 represents a completely opaque pixel, one you cannot see through
 - Value of 0 is a completely transparent pixel
 - $0 < \alpha < 1 \Rightarrow$ translucency
- Useful for blending images
 - Images with higher alpha values are less transparent
 - Linear interpolation ($\alpha X + (1 - \alpha)Y$) or full Porter-Duff compositing algebra

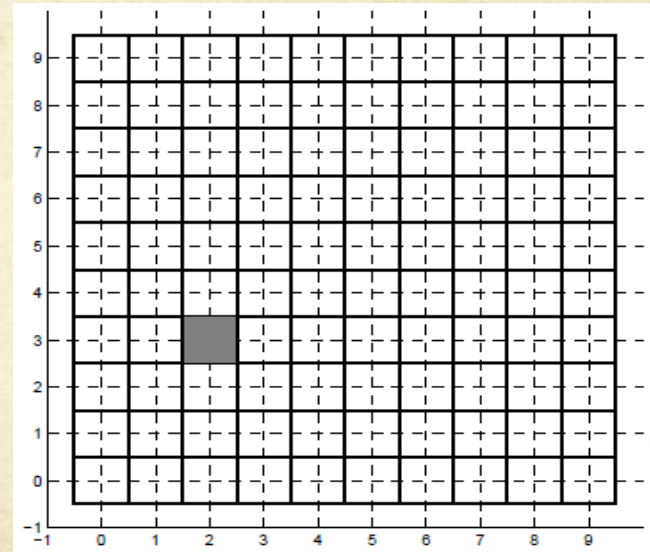


The orange box is drawn on top of the purple box using $\alpha = 0.8$

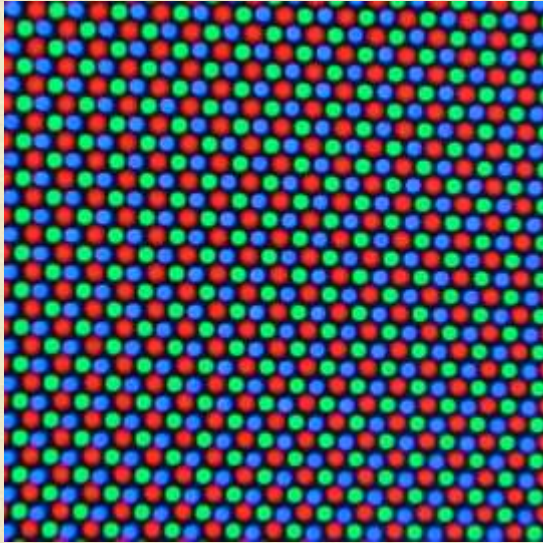
3

Modeling an image

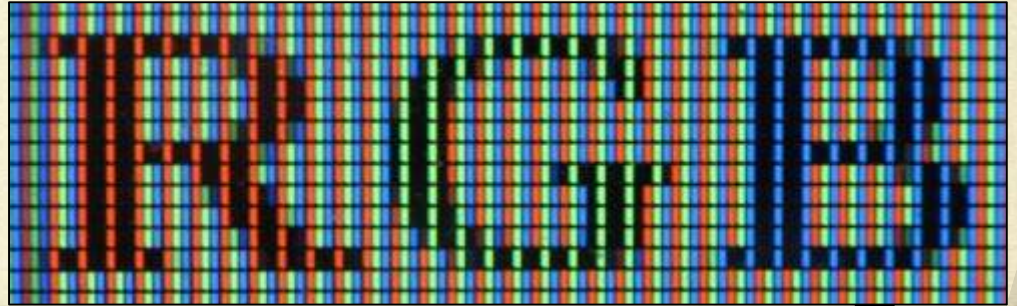
- Model a one-channel image as the function from pairs of integers to real numbers
 - x and y are integers such that x and y
- Associate each pixel value to small area around display location with coordinates
- A pixel here looks like a square centered over the sample point, but it's just a scalar value and the actual geometry of its screen appearance varies by device
 - Roughly circular spot on CRT
 - Rectangular on LCD panel



- Pixels are point samples, not “squares” or “dots”
- Point samples reconstructed for display (often using multiple subpixels for primary colors)



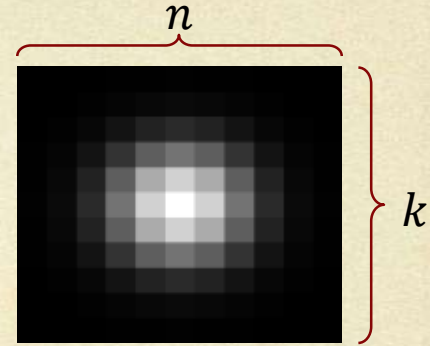
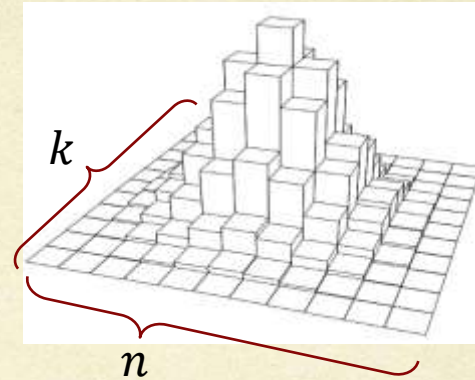
Close-up of a CRT screen



Close-up of an LCD screen

Discrete Images vs. Continuous Images

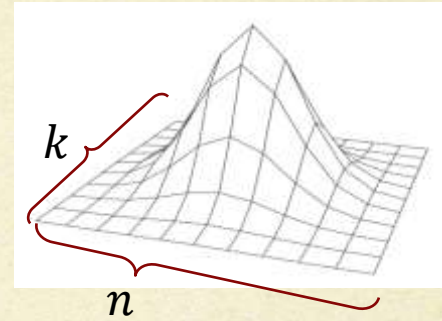
- Two kinds of images
 - Discrete
 - Continuous
- Discrete image
 - Function from $\mathbb{Z}^2$ to $\mathbb{R}$
 - How images are stored in memory
 - The kind of images we generally deal with as computer scientists



Discrete image $u(i, j)$

Discrete Images vs. Continuous Images

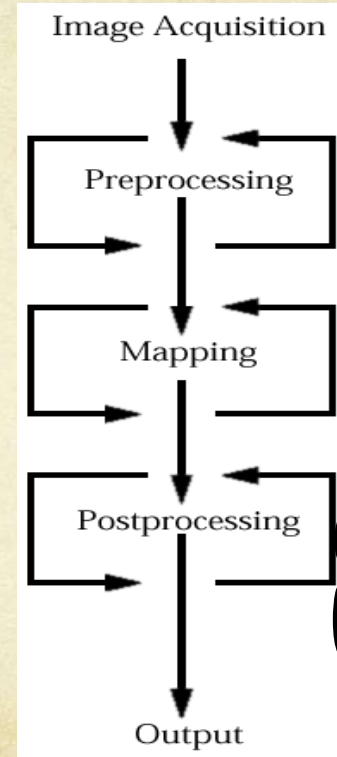
- Continuous image
 - Function from $\mathbb{R}^2$ to $\mathbb{R}$
 - Images in the real world
 - “Continuous” refers to the domain, not the values (discontinuities could still exist)
- Example: Gaussian distribution
 - i and j are the center of the Gaussian
 - i ,
 - j ,
 - and n (n odd)
 - Here i and j



Continuous image $v(i, j)$

Idealized Five Stage Pipeline of Image Processing

- The stages are
 - Image acquisition - how we obtain images in the first place
 - Preprocessing - any effects applied before mapping (e.g., crop, mask, filter)
 - Mapping - catch-all stage involving image transformations or image composition
 - Postprocessing - any effects applied after mapping (e.g., texturizing, color remapping)
 - Output - printing or displaying on a screen
- In practice, stages may be skipped
- Middle stages are often interlaced



Stage 1: Image Acquisition

- Image Synthesis
 - Images created by a computer
 - Painted in 2D
 - Corel Painter ([website](#))
 - Photoshop ([website](#))
 - Rendered from 3D geometry
 - Pixar's RenderMan ([website](#))
 - Autodesk's Maya ([website](#))
 - Your CS123 projects
 - Procedurally textured
 - Generated images intended to mimic their natural counterparts
 - e.g., procedural wood grain, marble
- Image Capture
 - Images from the “real world”
 - Information must be digitized from an analog signal
 - Common capture methods:
 - Digital camera
 - Satellite data from sensors (optical, thermal, radiation,...)
 - Drum scanner
 - Flatbed photo scanner
 - Frames from video

Stage 2: Preprocessing

- Each source image is adjusted to fit a given tone, size, shape, etc., to match a desired quality or to match other images
- Can make a set of dissimilar images appear similar (if they are to be composited later), or make similar parts of an image appear dissimilar (such as contrast enhancement)



Original



Adjusted grayscale curve

Stage 2: Preprocessing (continued)

- Preprocessing techniques include:

- Adjusting color or grayscale curve



- Cropping



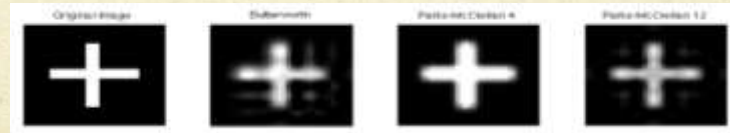
- Masking (cutting out part of an image)



- Blurring and sharpening



- Edge detection/enhancement



- Filtering and antialiasing



- Scaling up (super sampling) or scaling down (sub sampling)



Stage 2: Preprocessing (continued)

- Notes:
 - Blurring, sharpening, and edge detection can also be postprocessing techniques
 - Some preprocessing algorithms are not followed by mapping, others that involve resampling the image may be interlaced with mapping: filtering is done this way

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Stage 3: Mapping

- Mapping is a catch-all stage where several images are combined, or geometric transformations are applied
- Transformations include:
 - Rotating
 - Scaling
 - Shearing
 - Warping
 - Feature-based morphing
- Compositing:
 - Basic image overlay
 - Smooth blending with alpha channels
 - Poisson image blending
 - Seamlessly transfers “details” (like edges) from part of one image to another

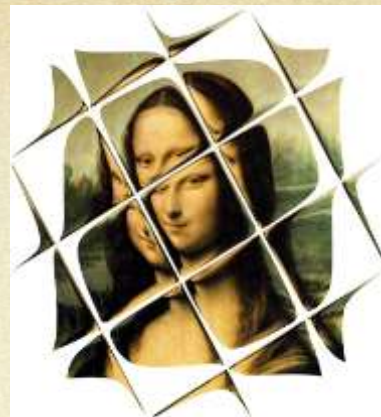


Image Warping



Poisson Image
Blending

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Stage 4: Postprocessing

- Creates global effects across an entire image or selected area
- Art effects
 - Posterizing
 - Faked “aging” of an image
 - Faked “out-of-focus”
 - “Impressionist” pixel remapping
 - Texturizing
- Technical effects
 - Color remapping for contrast enhancement
 - Color to B&W conversion
 - Color separation for printing (RGB to CMYK)
 - Scan retouching and color/contrast balancing



Posterizing



Aging



Impressionist

Stage 5: Output (Archive/Display)

- Choice of display/archive method may affect earlier processing stages
 - Color printing accentuates certain colors more than others
 - Colors on the monitor have different **gamuts** and **HSV** values than the colors printed out
 - Need a mapping
 - HSV = hue, saturation, value, a cylindrical coordinate system for the RGB color model
 - Gamut = set of colors that can be represented by output device/printer
- Display Technologies
 - Monitor (CRT → LCD/LED/OLED/Plasma panel)
 - Color printer
 - Film/DVD
 - Disk file
 - Texture map for 3D renderer

Example 1: Edge Detection Filtering

- Edge detection filters measure the difference between adjacent pixels
- A greater difference means a stronger edge
- A threshold is sometimes used to remove weak edges

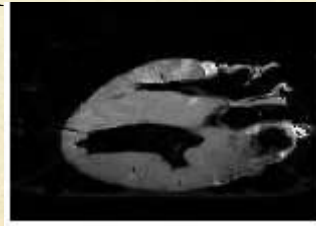


*Sobel edge
detection filter* →



Example 1: Edge Detection Filtering (Continued)

- Used with MRI scans to reveal boundaries between different types of tissues
- MRI scan is image where gray level represents tissue density
- Used same filter as previous slide



Original MRI
image
of a dog heart

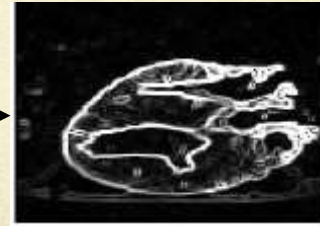
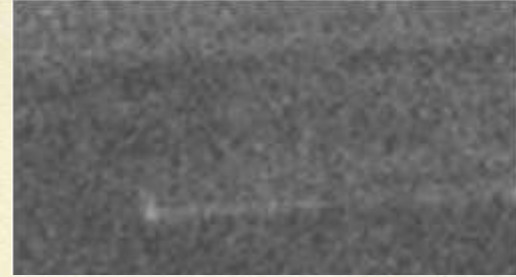


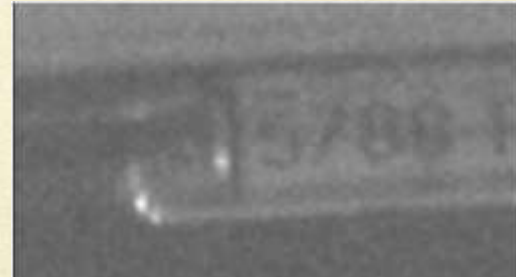
Image after
edge detection

Example 2: Image Enhancement for Forensics

- Extract evidence from seemingly incomprehensible images
- Normally, image enhancement uses many filtering steps, and often no mapping at all
- Michael Black and his class, CS296-4, received a commendation for helping Virginia police in a homicide case



Before enhancement



After enhancement

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Example 2: Image Enhancement for Forensics

- We have a security camera video of the back of a car that was used in a robbery
- The image is too dark and noisy for the police to pull a license number
- Though humans can often discern an image of poor quality, filtering can make it easier for a pattern-recognition algorithm to decipher embedded symbols
 - Optical Character Recognition
- Step 1: Get the frame from the videotape digitized with a frame-grabber



Example 2: Image Enhancement for Forensics

- Step 2: Crop out stuff that appears to be uninteresting (outside plate edges)
- This step can speed process by doing image processing steps on fewer pixels
- Can't always be done, may not be able to tell which sections are interesting without some processing



Example 2: Image Enhancement for Forensics

- Step 3: Use edge-sharpening filter to add contrast to plate number
- This step enhances edges by raising discontinuities at brightness gaps in image



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Example 2: Image Enhancement for Forensics

- Step 4: Remap colors to enhance contrast between numbers and plate itself
- Now, can make a printout for records, or just copy plate number down: YNN-707!
- Note that final colors do not even resemble real colors of license plate—enhancement techniques have seriously distorted the colors!



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Multipart Composition

- Image composition is popular in the art world, as well as in tabloid news
- Takes parts of several images and creates single image
 - Hard part is making all images fit together naturally
- Artists can use it to create amazing collages and multi-layered effects
- Tabloid newspaper artists can use it to create “News Photos” of things that never happened – “Fauxtography”.
 - There is no visual truth in media!

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Famous Faked Photos

Fonda Speaks To Vietnam Veterans At Anti-War Rally



Actress And Anti-War Activist Jane Fonda Speaks to a crowd of Vietnam Veterans as Activist and Former Vietnam Vet John Kerry (L, SFT) listens and prepares to speak next concerning the war in Vietnam (AP Photo)



Chinese press photo of
Tibet railway



Tom Hanks and JFK

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Example image composition (1/5)

- Lars Bishop, former CS123 Head TA, created a news photo of himself “meeting” with former Russian President Boris Yeltsin
 - post-Gorbachev and Perestroika. He served 10 July 1991 – 31 December 1999, resigned in favor of Putin)
- Needless to say, Lars Bishop never met Mr. Yeltsin
- Had to get the images, cut out the parts he wanted, touch them up, paste them together, and retouch the end result



Image of Boris (from Internet)



Image of Lars (from video camera)

Example image composition (2/5)

- Cut the pictures we want out of the original images
- Paint a region around important parts of images (outline of people) using Photoshop
- Continue touching up this outline until no background at edge of people
- Use a smart lasso tool that grows until it hit the white background, thus selecting subject. (“Magic Wand” tool in photoshop can accomplish this)



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Example image composition (3/5)

- Filter the images to make them appear similar, and paste them together
 - Boris is blurred and brightened to get rid of the halftoning lines (must have been a magazine photo)
 - Lars is blurred and noise is added to match image quality to that of Boris
 - Images are resized so Boris and Lars are at similar scales



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Example image composition (4/5)

- Finalize image
 - Created a simple, two-color background and added noise so it fit with the rest of the image, placed cutout of the two subjects on top of background
 - This left a thin white halo around the subjects, so used a “Rubber Stamp” tool to stamp background noise patterns over halo, making seams appear less obvious

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Example image composition (5/5)

- Final Image (with retouching at edges)



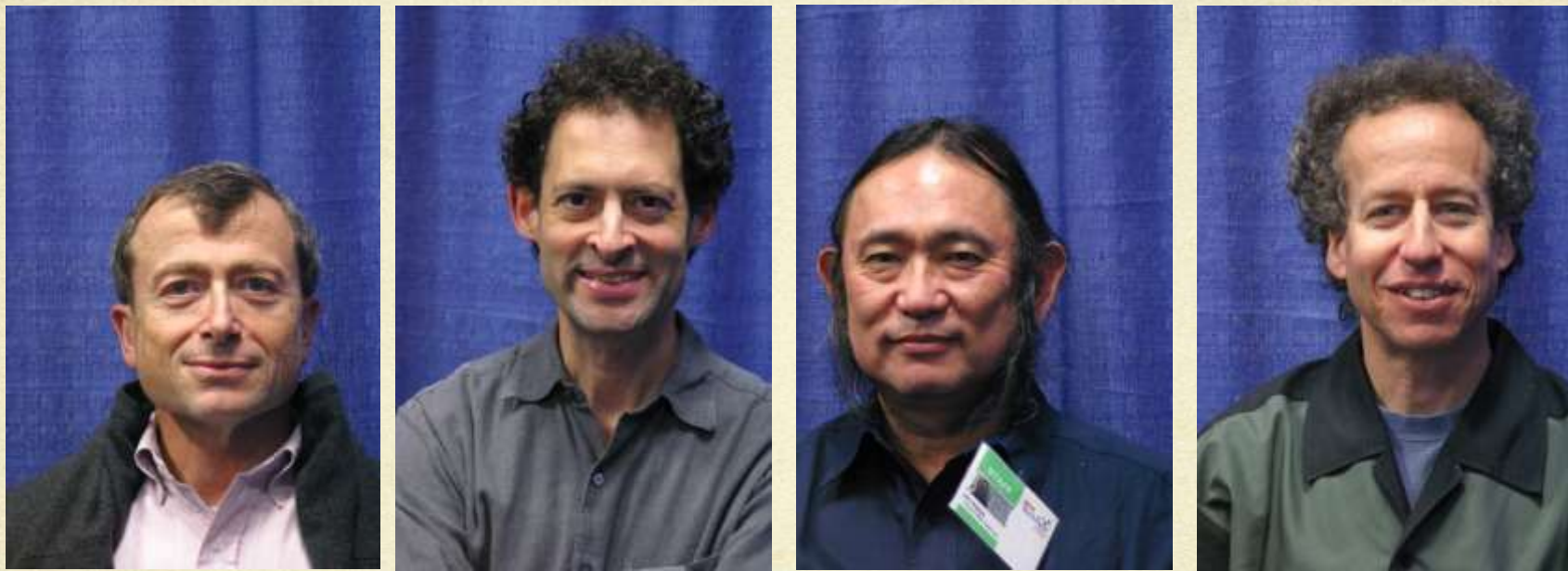
BISHOP AND YELTSIN TALK PEACE

BISHOP: "I couldn't understand a single word he said!"

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Image Composition – Frankenface



Aseem Agarwala, Mira Dontcheva, Maneesh Agrawala, Steven Drucker, Alex Colburn, Brian Curless, David Salesin, Michael Cohen. **Interactive Digital Photomontage**. *ACM Transactions on Graphics (Proceedings of SIGGRAPH 2004)*, 2004.

<http://grail.cs.washington.edu/projects/photomontage/>

Image Composition – Frankenface



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<http://grail.cs.washington.edu/projects/photomontage/>

Computer Vision (1/2)

- Computer graphics is the business of using models to create images; computer vision solves the opposite problem—deriving models from images
- Computer must do all the processing without human intervention
- Often, processing techniques must be fast
 - Slow processing will add to camera-to-reaction latency (lag) in system
- Common preprocessing techniques for computer vision:
 - Edge enhancement
 - Region detection
 - Contrast enhancement
 - Feature point detection

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Computer Vision (2/2)

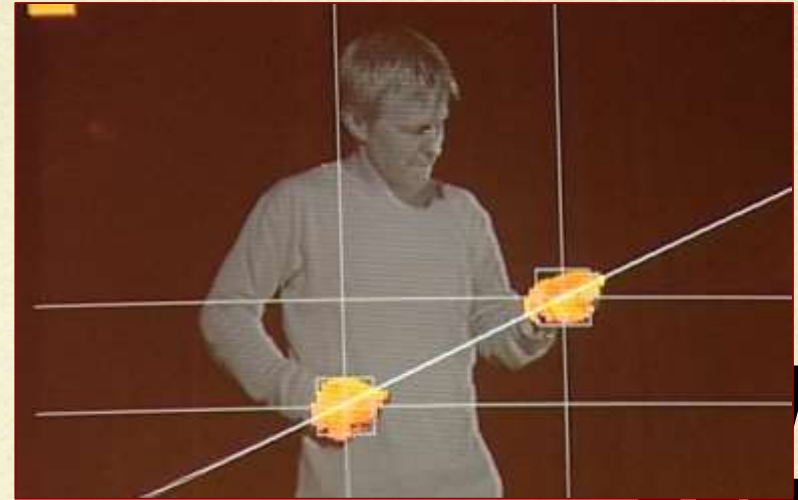
- Image processing makes information easier to find
- Pattern detection and pattern recognition are separate fields in their own right
 - Pattern detection: looking for features and describing the image's content at a higher level
 - Pattern recognition: classifying collections of features and matching them against library of stored patterns. (e.g., alphanumeric characters, types of abnormal cells, or human features in the case of biometrics)
 - Pattern detection is one important component of pattern recognition
- Computer vision can be used as part of a passive UI, as an alternative to intrusive (tethered) gadgetry such as 6DoF “space mice”, wands, and data gloves – see next slide
- Computational photography draws on many techniques from vision
- For more on computer vision: CS143 (computer vision), CS129 (computational photography). Also courses in CLPS, ENGN

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Example: Virtual Air Guitar

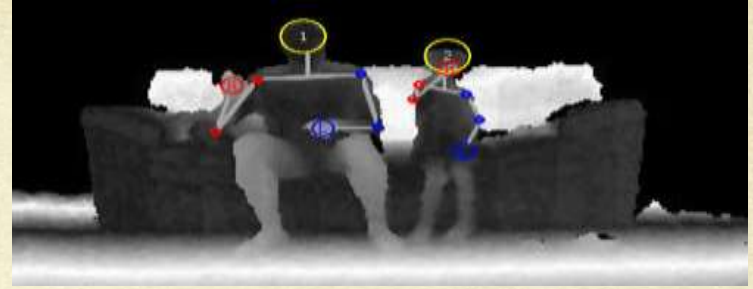
- Real-time computer vision system
 - User wears orange gloves
 - Glove “regions” are extracted in real-time
- Gesture recognition
 - Change note with upper hand
 - Strum with lower hand
 - Whenever hand crosses virtual guitar neck (diagonal line in picture)
 - Also vibrato and slide gestures



The Virtual Air Guitar from
Helsinki University of Technology

Microsoft Kinect

- Uses computer vision to “see” your body’s shape
 - Extract multiple “skeletons” from depth image
 - Body as a controller
 - Gesture recognition
 - Facial recognition
- Works with cheap hardware
 - RGB camera
 - CMOS depth sensor
 - Projected infrared pattern to see in darkness
 - Total cost around \$100
- Current research uses Kinects to construct 3D models
 - [Kinect Fusion](#)



Joints of skeletons on top of depth map



Output from Kinect Fusion.
This is an example of model capture



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3D Image Processing: (1/3)

- 3D images
 - 3D image volumes from MRI scans need image processing
 - 2D image processing techniques often have 3D analogs
 - Display becomes more difficult: **voxels** replace pixels (**volumetric rendering**)
 - Increases time and space complexity:
 - 4 channel 1024x1024 image = 4 megs
 - 4 channel 1024x1024x1024 image = 4 **gigs!**
 - processing algorithms become

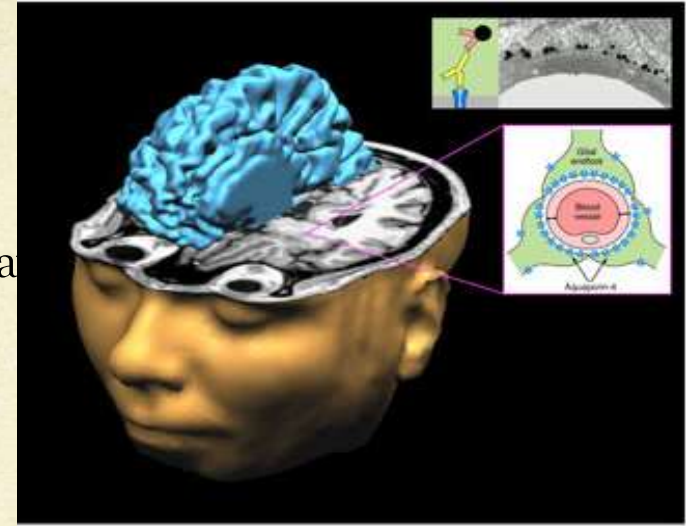
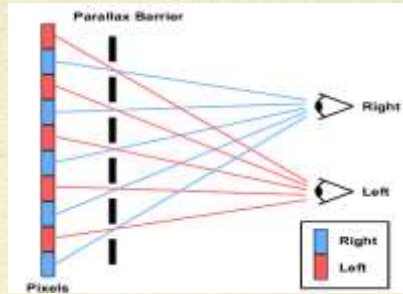


Illustration: Erlend Nagelhus and Gunnar Leikanger
3D MRI: Kyrre Eeg Emblem, Rikshospitalet, and Inge Rasmussen, Nidelven Hjerneforskningslaboratorium.

University of Oslo, 1999

3D Image Processing: (2/3)

- 3D displays
 - Autostereoscopic displays starting to emerge (no glasses or other headgear)
 - Most work with lenticular optics or parallax effects
 - We'll revisit this in more detail later in the semester
 - Several companies are commercializing products
 - Philips released the first 3D HDTV in 2009, but 3D TVs haven't had great commercial impact thus far
 - Panasonic, LG, Samsung and Sony also have 3D TVs on the market
 - Hitachi, HTC, LG all produce 3D mobile phones
 - Nintendo 3DS



HTC EVO 3D



Nintendo 3DS

3D Image Processing: (3/3)

- zSpace, an interactive 3D display and computing platform
 - <http://zspace.com/zspace-system/>
 - Potential applications include architecture, data visualization, medicine, digital art, engineering, gaming/entertainment, education
- Fujifilm 3D technology
 - FinePix REAL 3D, released in 2009, captures video and images in 3D
 - 3D printing service



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